

MATH 2211, SPRING 2026: SOLUTIONS TO HOMEWORK 6

- (1) Suppose that we have a matrix $A \in M_{n \times m}(F)$ associated with a linear transformation $T : F^m \rightarrow F^n$. How would you use row reduction to verify the following:
- If T is injective?
 - If T is surjective?
 - (when $m = n$) If T is an invertible linear transformation? In this case, also give a method for finding the inverse matrix A^{-1} (see Lecture 17).

Solution. A linear transformation $T : F^m \rightarrow F^n$ is injective if and only if $\ker T = \{0\}$; see the proof of the Observation from Lecture 19. So to check injectivity, we take A , put it into RREF, and verify that there are no free variables. In other words, we check that every column has a leading 1.

It is surjective if and only if $\text{im } T$ is all of F^n . This is the same as saying that $\dim \text{im } T = n$. In turn, this is equivalent to saying that we have n leading 1s, which is equivalent to saying that every *row* of A has a leading 1.

The same Observation from Lecture 19 shows that T is invertible precisely when the RREF of A is I_n .

The process for finding the inverse matrix A^{-1} is now as follows: The columns of the matrix are $A^{-1}e_i$ for $i = 1, \dots, n$. Solving for $A^{-1}e_i = v_i$ is the same as solving for $Av_i = e_i$ (why?). Therefore, the i -th column of A^{-1} is the solution to the system $[A \mid e_i]$. We now do row operations to get A into RREF and this brings us to the system $[I_n \mid v_i]$. In other words, here are the steps for finding the inverse:

- Consider the system $[A \mid I_n]$.
- Do row operations to get A into $\text{RREF}(A) = I_n$.
- Do the *same* row operations to I_n to end up with the system $[I_n \mid A^{-1}]$

- (2) Show that the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 3 & -1 & 0 \\ 1 & 0 & 2 \end{bmatrix} \in M_{n \times n}(\mathbb{Q})$$

is invertible and find its inverse. Use the inverse to solve the equations

$$Ax = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} ; Ax = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Solution. We do our row operations simultaneously to A and I_3 :

$$\begin{aligned}
 \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 3 & -1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 1 \end{array} \right] & \xrightarrow[\substack{R2-3\cdot R1 \\ R3-R1}]{} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -4 & -3 & -3 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array} \right] \\
 & \xrightarrow{R2 \leftrightarrow R3} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \\ 0 & -4 & -3 & -3 & 1 & 0 \end{array} \right] \\
 & \xrightarrow{(-1)\cdot R2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & -1 \\ 0 & -4 & -3 & -3 & 1 & 0 \end{array} \right] \\
 & \xrightarrow[\substack{R3+4\cdot R2 \\ R1-R2}]{} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 & 0 & -1 \\ 0 & 0 & -7 & 1 & 1 & -4 \end{array} \right] \\
 & \xrightarrow{(-7)^{-1}\cdot R3} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 & 0 & -1 \\ 0 & 0 & 1 & -\frac{1}{7} & -\frac{1}{7} & \frac{4}{7} \end{array} \right] \\
 & \xrightarrow[\substack{R2+R3 \\ R1-2\cdot R3}]{} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{2}{7} & \frac{2}{7} & -\frac{1}{7} \\ 0 & 1 & 0 & \frac{6}{7} & -\frac{1}{7} & -\frac{3}{7} \\ 0 & 0 & 1 & -\frac{1}{7} & -\frac{1}{7} & \frac{4}{7} \end{array} \right]
 \end{aligned}$$

(3) Consider the function

$$\begin{aligned}
 T : \text{Poly}_{\leq 3}(F) &\longrightarrow \text{Poly}_{\leq 3}(F) \\
 f(x) &\mapsto f(0)x^3 + f(1)x^2 + f(1)x + f(0)
 \end{aligned}$$

Show that T is a linear transformation and compute its matrix with respect to the bases

$$\mathcal{B} = (1, x, x^2, x^3); \mathcal{B}' = (1, (x-1), (x-1)^2, (x-1)^3).$$

Solution. Checking that this is a linear transformation just amounts to seeing that $(af_1 + bf_2)(c) = af_1(c) + bf_2(c)$ for all $a, b, c \in F$ and $f_1(x), f_2(x) \in \text{Poly}_{\leq 3}(F)$.

To compute its matrices, we check:

$$\begin{aligned}
 T(1) &= x^3 + x^2 + x + 1 = 1 \cdot 1 + 1 \cdot x + 1 \cdot x^2 + 1 \cdot x^3 \\
 T(x) &= x^2 + x = 0 \cdot 1 + 1 \cdot x + 1 \cdot x^2 + 0 \cdot x^3 \\
 T(x^2) &= x^2 + x = 0 \cdot 1 + 1 \cdot x + 1 \cdot x^2 + 0 \cdot x^3 \\
 T(x^3) &= x^2 + x = 0 \cdot 1 + 1 \cdot x + 1 \cdot x^2 + 0 \cdot x^3
 \end{aligned}$$

Therefore, we find

$$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

For the other basis, we use the Taylor expansions of polynomials at 1

$$\begin{aligned} T(1) &= x^3 + x^2 + x + 1 = 4 \cdot 1 + 6 \cdot (x-1) + \frac{8}{2} \cdot (x-1)^2 + \frac{6}{3!} \cdot (x-1)^3 \\ T(x-1) &= -x^3 - 1 = (-2) \cdot 1 + (-3) \cdot (x-1) - \frac{6}{2} \cdot (x-1)^2 - \frac{6}{3!} \cdot (x-1)^3 \\ T((x-1)^2) &= x^3 + 1 = 2 \cdot 1 + 3 \cdot (x-1) + \frac{6}{2} \cdot (x-1)^2 + \frac{6}{3!} \cdot (x-1)^3 \\ T((x-1)^3) &= -x^3 - 1 = 2(-2) \cdot 1 + (-3) \cdot (x-1) - \frac{6}{2} \cdot (x-1)^2 - \frac{6}{3!} \cdot (x-1)^3 \end{aligned}$$

Therefore,

$$[T]_{\mathcal{B}'} = \begin{bmatrix} 4 & -2 & 2 & -2 \\ 6 & -3 & 3 & -3 \\ 4 & -3 & 3 & -3 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

(4) Suppose that V and W are finite dimensional F -vector spaces, and define

$$\text{Hom}_F(V, W) = \{F\text{-linear transformations } T : V \rightarrow W\}$$

(the Hom is short for homomorphism).

(a) Show that $\text{Hom}_F(V, W)$ is an F -vector space with linear combinations given by:

$$a_1 T_1 + a_2 T_2 : v \mapsto a_1 T_1(v) + a_2 T_2(v) \in W$$

for $a_1, a_2 \in F$ and $T_1, T_2 \in \text{Hom}_F(V, W)$.

Solution. Set $T = a_1 T_1 + a_2 T_2$. Seeing that this is once again a linear transformation amounts to checking for $b_1, b_2 \in F, v_1, v_2 \in V$, we have

$$\begin{aligned} T(b_1 v_1 + b_2 v_2) &= (a_1 T_1 + a_2 T_2)(b_1 v_1 + b_2 v_2) \\ &= a_1 T_1(b_1 v_1 + b_2 v_2) + a_2 T_2(b_1 v_1 + b_2 v_2) \\ &= a_1 T_1(b_1 v_1) + a_1 T_1(b_2 v_2) + a_2 T_2(b_1 v_1) + a_2 T_2(b_2 v_2) \\ &= a_1 b_1 T_1(v_1) + a_2 b_1 T_2(v_1) + a_1 b_2 T_1(v_2) + a_2 b_2 T_2(v_2) \\ &= b_1(a_1 T_1 + a_2 T_2)(v_1) + b_2(a_1 T_1 + a_2 T_2)(v_2) \\ &= b_1 T(v_1) + b_2 T(v_2). \end{aligned}$$

We have to check that the operation $(T_1, T_2) \mapsto T_1 + T_2$ is associative, which follows from associativity of addition in W . The zero element is of course given by the zero or trivial linear transformation $0 : v \mapsto 0$ for all $v \in V$. We also have to check that we have additive inverses: The additive inverse for T is given by the linear transformation $-T = (-1) \cdot T$. Finally, it is clear that $1 \cdot T = T$, and that we have $a_1(a_2 T) = (a_1 a_2)T$ for all $a_1, a_2 \in F$.

(b) If $\mathcal{B}_1$ and $\mathcal{B}_2$ are bases for V and W , respectively, with $n = \dim V$ and $m = \dim W$ show that the function

$$\begin{aligned} \alpha : \text{Hom}_F(V, W) &\rightarrow M_{n \times m}(F) \\ T &\mapsto [T]_{\mathcal{B}_1, \mathcal{B}_2} \end{aligned}$$

is an isomorphism of F -vector spaces.

Solution. We have to check that this is a bijective linear transformation. The quickest way to see this is to use the original definition of $[T]_{\mathcal{B}_1, \mathcal{B}_2}$: The bases $\mathcal{B}_1$ and $\mathcal{B}_2$ correspond to isomorphisms

$$U_1 : F^n \xrightarrow[e_i \mapsto v_i]{\cong} V ; U_2 : F^m \xrightarrow[e_j \mapsto w_j]{\cong} W.$$

The matrix $[T]_{\mathcal{B}_1, \mathcal{B}_2}$ now represents the linear transformation

$$U_2^{-1} \circ T \circ U_1 : F^n \rightarrow F^m.$$

The fact that $T \mapsto [T]_{\mathcal{B}_1, \mathcal{B}_2}$ is a linear transformation now corresponds to the fact that, for $a_1, a_2 \in F$ and $T_1, T_2 \in \text{Hom}_F(V, W)$, we have

$$U_2^{-1} \circ (a_1 T_1 + a_2 T_2) \circ U_1 = a_1 (U_2^{-1} \circ T_1 \circ U_1) + a_2 (U_2^{-1} \circ T_2 \circ U_1).$$

This uses primarily the fact that U_2^{-1} is a linear transformation (why?).

Conversely, if we have a matrix $A \in M_{m \times n}(F)$ corresponding to the linear transformation $S : F^n \xrightarrow{e_i \mapsto Ae_i} F^m$, then the linear transformation $T = U_2 \circ S \circ U_1^{-1} : V \rightarrow W$ satisfies $[T]_{\mathcal{B}_1, \mathcal{B}_2} = A$. This gives an inverse to the assignment $T \mapsto [T]_{\mathcal{B}_1, \mathcal{B}_2}$ and shows that it is bijective.

- (c) What is the dimension of $\text{Hom}_F(V, W)$?

Solution. Since $M_{n \times m}(F)$ has dimension nm (why?), we see that $\text{Hom}_F(V, W)$ must also have the same dimension.

An $n \times n$ -matrix is **upper triangular** if it doesn't have any non-zero entries below the diagonal. That is, if $a_{i,j} = 0$ whenever $i > j$. An F -linear transformation $T : V \rightarrow V$ on a finite dimensional vector space V is **upper triangulizable** if there exists a basis $\mathcal{B}$ for V such that $[T]_{\mathcal{B}}$ is upper triangular.

- (5) Show that the following are equivalent for $T : V \rightarrow V$

- (a) T is upper triangulizable.
- (b) There exists a basis $\{v_1, v_2, \dots, v_n\}$ of V such that, for each j ,

$$T(v_j) = \sum_{i \leq j} b_{i,j} v_i,$$

for some $b_{i,j} \in F$.

- (c) There exists a basis $\{v_1, v_2, \dots, v_n\}$ of V such that, for each i ,

$$T(\text{span}(v_1, \dots, v_i)) \leq \text{span}(v_1, \dots, v_i).$$

Solution. (1) $\Leftrightarrow$ (2): Given the definition of $[T]_{\mathcal{B}}$, this is just restating the definition with $a_{i,j}$ replaced by $b_{i,j}$.

(2) $\Leftrightarrow$ (3): This arises from the fact that both statements are equivalent to saying that, for each j , $T(v_j)$ is a F -linear combination of $v_1, \dots, v_j$.